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Drug Adverse Side Effect Predictions and Doctor Assistance

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ABSTRACT- Adverse Drug Reactions (ADRs) continue to be one of the major challenges in healthcare, even though pharmaceutical science has advanced significantly in recent decades. Adverse drug reaction varies from one patient to another, not because of the drug but upon biological, genomic, and lifestyle attributes of the patients. Earlier, classical pharmacovigilance did not get forward to defining these in patients or health professionals in an event-response mode. The study anticipates a system that will be based on a completely new paradigm, predicting possible side-effects before usage.

RAG holds for system architecture and works with datasets such as SIDER and FAERS in providing patient-specific drug risk assessment. It is an assessment layer for enhancement of clinical fidelity. The system is also furnished with adverse effects wellness recommendations through Ayurvedic remedies and lifestyle modifications- bearing in mind the growing trend towards holistic treatment. The system does not have a target of eliminating ADs, which is far from being possible, but rather of reducing their number to the maximum possible extent and aiding clinicians in their decisions.

KEYWORDS- Adverse Drug Reactions, RAG, AI in Healthcare, Doctor-in-the-Loop, Fast API, Vector Search, Patient Safety.

INTRODUCTION

Background and Rationale

Modern medicine has a myriad of benefits: improving life expectancy for millions, raising the quality of life for many, and so on. However, such an upsurge created an entirely new chapter of problems that modern medicine now faces. One of these new problems is the increasing number of 'ADRs' or adverse drug reactions, which result from inadvertent side effects from using medicines at conventional dosage levels. All health organizations worldwide hold that adverse effects account for multi-hospital admissions, and they are among the leading causes of adverse, preventable, medical events.

The array of drugs stacked up is quickly growing along with variant patient populations. What works for one person can very much throw out another with severe complications, such as genetic makeup, comorbidity, age, or any lifestyle-related factor such as diet or sleep. Probably the greatest risk presented by almost all patients is the possibility of polypharmacy. Double-checking all medications by hand becomes practically impossible in most practical terms.

Existing Limitations in ADR Detection

Most of the present paradigms of pharmacovigilance have post-marketing and post-event these effects. Adverse effect is only reported after the patient is exposed to it. Meaning, injury has now occurred. It's very frequently outpaced by new research, drug interaction updates, and new reports that come to light.

The reality of the environment is that it certainly creates a need for tools that would be capable of predicting ADRs in a more personalized and also timely fashion. Innovations have been achieved in artificial intelligence, but the application has hardly sped up in practice of true clinical environments. This is mainly because clinicians are still skeptic of algorithmic predictions devoid of human supervision.

Motivation Behind the Study

Thus, we designed our system in such a way that it does not completely replace the doctor nor run entirely to the AI but strikes a very nice balance-very quickly, given the nature by AI processing huge amounts of data, followed by a final assessment by clinicians. Of course, study would lend itself well to this hybrid model: really the best of both worlds-the speed and all-encompassing nature of